

Opening AI the Data-centric way:

The Case for Data Institutions in Democratising the Design & Development of ML Datasets

THE PROPOSITION

Establishing dedicated data Institutions to steward ML datasets leverages datasets as strategic policy objects to create public value and insert a decentralised, community-accountable governance layer into the ML development process.

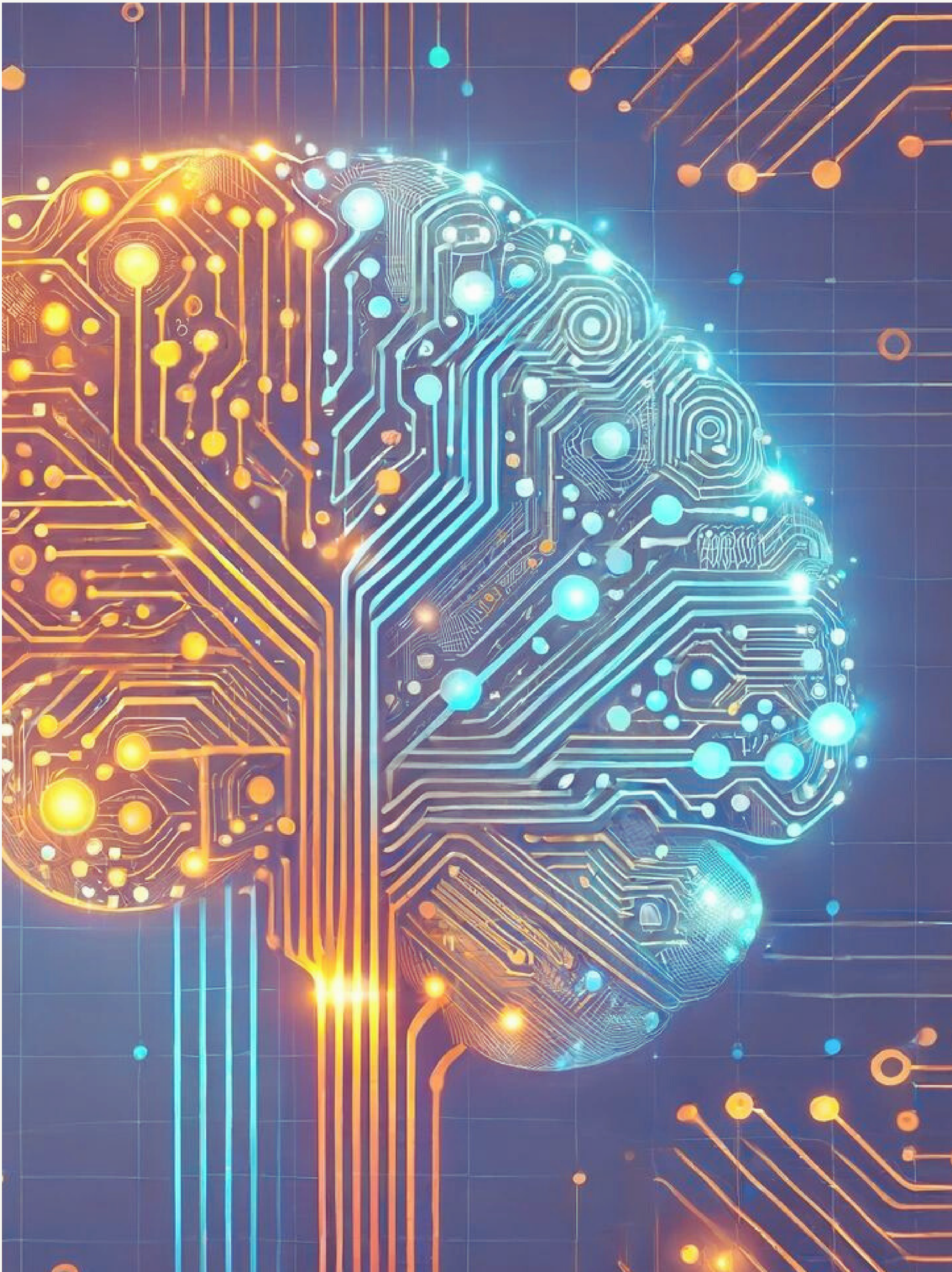
Data institutions offer a mechanism for democratic accountability and a policy lever to address current (mal)practice in dataset development around data quality, usage, sustainability, and embedding of good governance practices.

Further, dedicated data institutions for ML datasets close AI's paradox of openness, whereby AI developers exploit open source AI discourse and access to the digital commons to deploy publicly available knowledge, tools and data for training models which are then packaged into commercial products. Data institutions facilitate the return of value to those who generated the data/knowledge, and so contribute to the preservation and enrichment of the digital commons.

WHY DATASETS?

Datasets represent **strategic** policy objects within otherwise **opaque** ML value chains:

- Data enters AI systems through datasets; data is foundational to contemporary ML.
- Dataset production encodes and generates a variety of existing **harms** and biases.
- Datasets are **engineered** by stakeholders with interests through a series of definable processes; they reflect and shape normatively-desirable societal values.
- Benchmark datasets **set the agenda** and structure incentives for ML research.



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WHY DATA INSTITUIONS?

Data institutions represent a **broad family** of institutional templates, offering flexibility to meet divers contextual requirements. Bottom-up data institutions offer an extensible model of democratic data governance predicted on two core principles:

1. **Delegated fiduciary responsibility,**
2. **Collective or commons-based governance.**

This parsimonious focus makes them appropriate governance templates for ML datasets.

HOW DO DATA INSTITUTIONS CLOSE AI'S PARAODOX OF OPENNESS?

Data Quality

Delivering quality data is costly. Without adequate opportunities for scrutiny and accountability quality is currently under-allocated. Data institutions support quality-assurance and accountability regimes within dataset production. Quality is a context-dependent term, but broadly requires two characteristics, that a dataset is:

- 1) **Safety enhancing** (debiased, re-risked, well sampled, coherently licensed, consensual, audited),
- 2) **ML-ready** (methodically annotated, balanced across classes, low bias, feature rich, well strucutred).

Data Sustainability

ML developers require dependable supply chains providing up-to-date high-quality data at scale, continuously. Meeting this demand requires managing the **delicate balance** between openness and safeguarding the rights and interests of contributors.

By **realigning contribution and development incentives** closer to the needs of data-generating communities, data institutions help sustain a digital commons as the basis for further development and AI innovation.

Risk, Liability, Licensing & Consent

Like quality, risk management is under-provisioned by existing data markets. A network of delegated stewarding institutions empowered to manage and license data to commercial ML developers, would be well-positioned to provide **clarity and compliance over licensing options** for divers ML use cases, alongside **consent-gathering** and potentially other **liability guarantees**.

Further, DIs could support **opt-out mechanisms** for data subjects, by facilitating individuals' data-forgetting rights through promoting technical traceability, provenance documentation, and transparency processes.

Equity & Inclusion

DIs support the participatory action principle that no policy should be developed without the direct participation of affected communities: **"nothing about us, nothing without us"**. Through their representation mechanisms, data institutions embed equity and inclusion by design.

In a global south context, data institutions' community-orientation further support **digital sovereignty**, and move to close the democratic deficit at the heart of global AI governance.

DERIVED POLICY PRINCIPLES FOR DATA INSTITUTION DESIGN

#1. No one size fits all Heterogeneous data is governed by different, often overlapping, legal, contextual and custodial frameworks.	#4. Ensure fair work DIs should provide sites of accountability and coordination for guaranteeing investment in quality data work.	#7. Share as much as possible ...while respecting the decisions of data subjects. Follow both the CARE Principles and the FAIR principles.
#2. Be democratic Serve and represent a defined constituency, usually the data-generating community and impacted stakeholders.	#5. Transparency norms Invest meaningful resources into documentation to improve understanding of datasets' contextual validity.	#8. Sustainability Structure incentive mechanisms for data contribution, return value to producers and perhaps even offer support for data producers and stewards.
#3. Assume a task community role This requires an interventionist approach, whereby DIs actively acquire data representative of the population or task at hand.	#6. Smart quality Incorporate quality assurance and validation tools, as well as semantic and other functionality into a dataset-as-a-service offering.	#9 Leverage agency through technical enforcement Verification systems and proof-of-learning frameworks can ensure model developers are using data in an authorised and compliant manner.